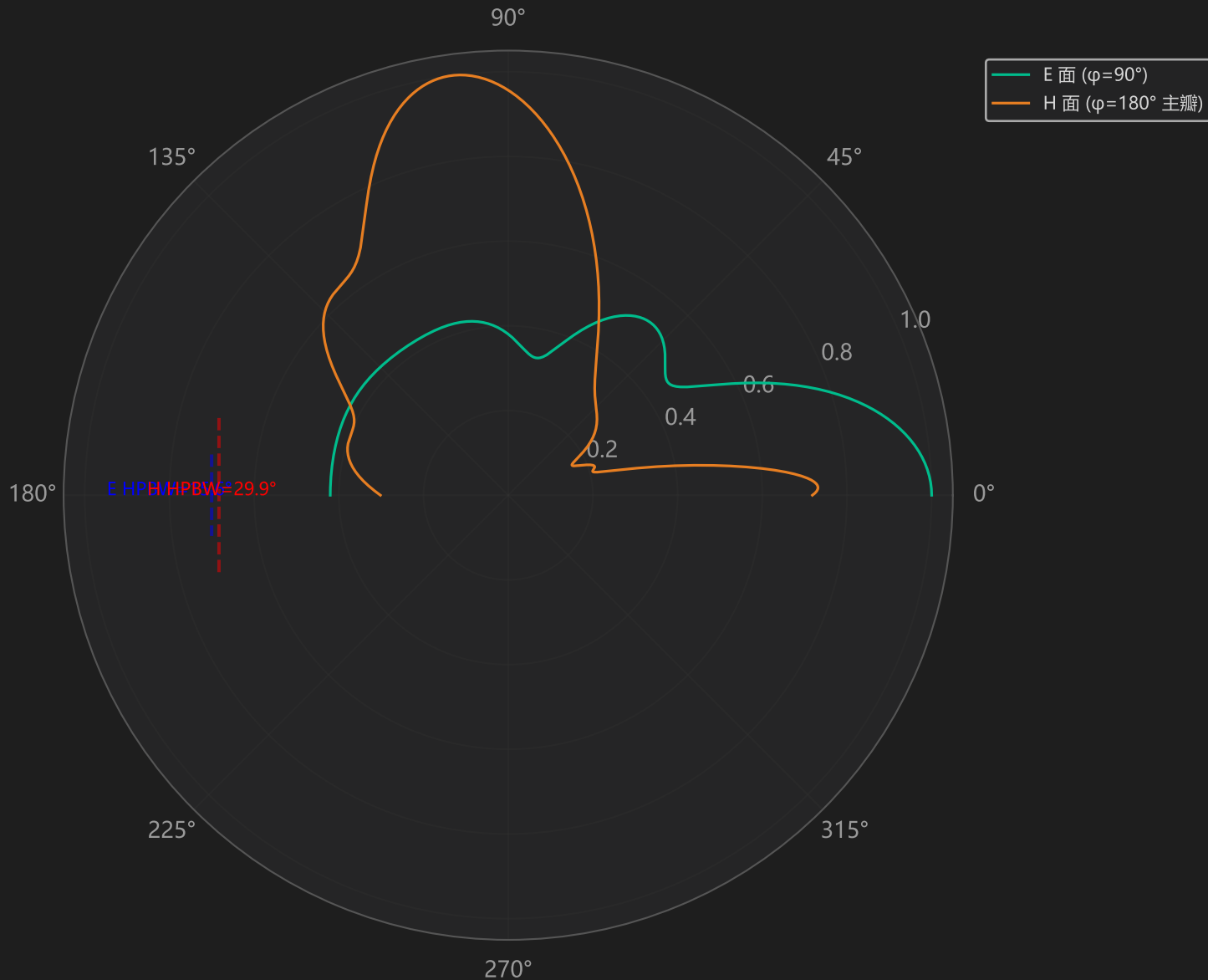
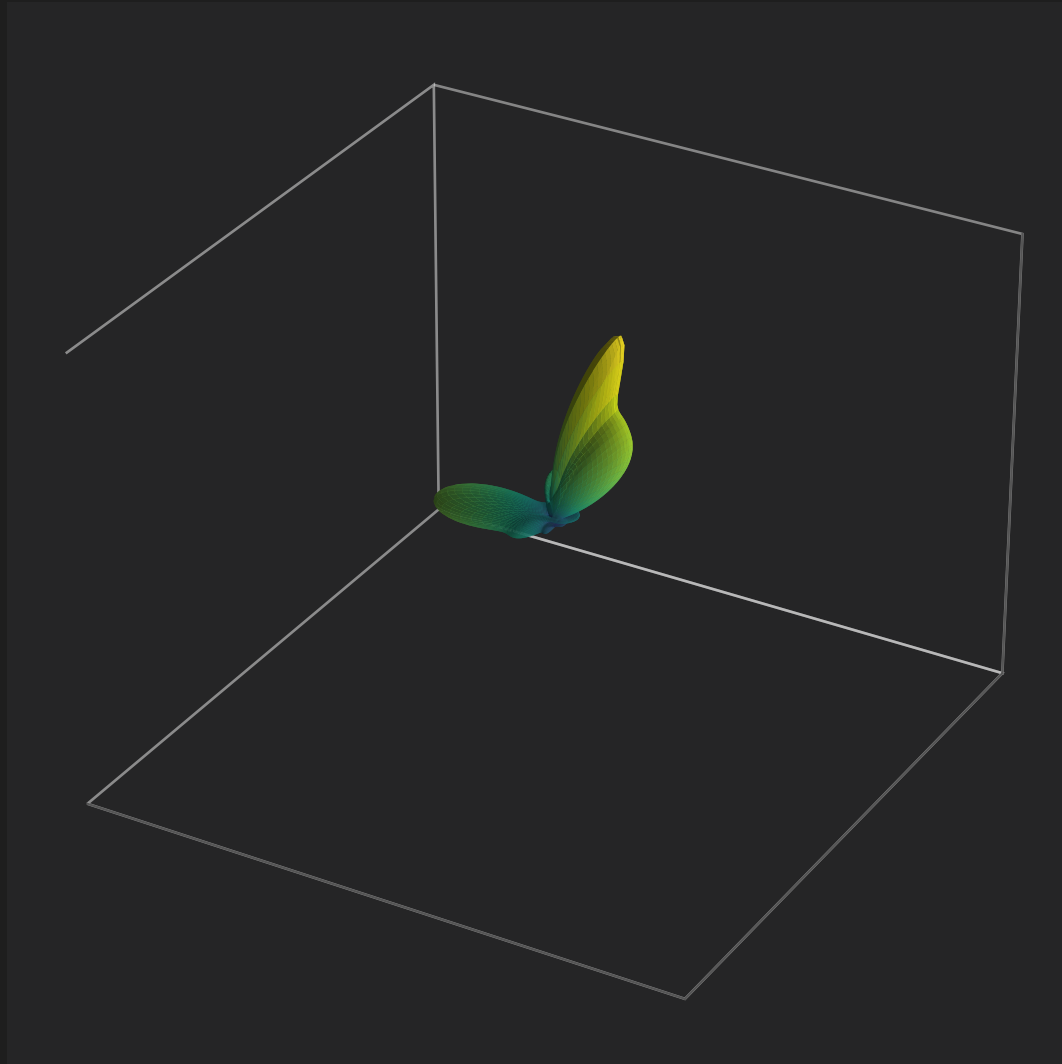


1420.0 MHz 天线仿真报告 — 半波长 ($\lambda/2$) | 抛物面 + 反射板 + 屏蔽桶

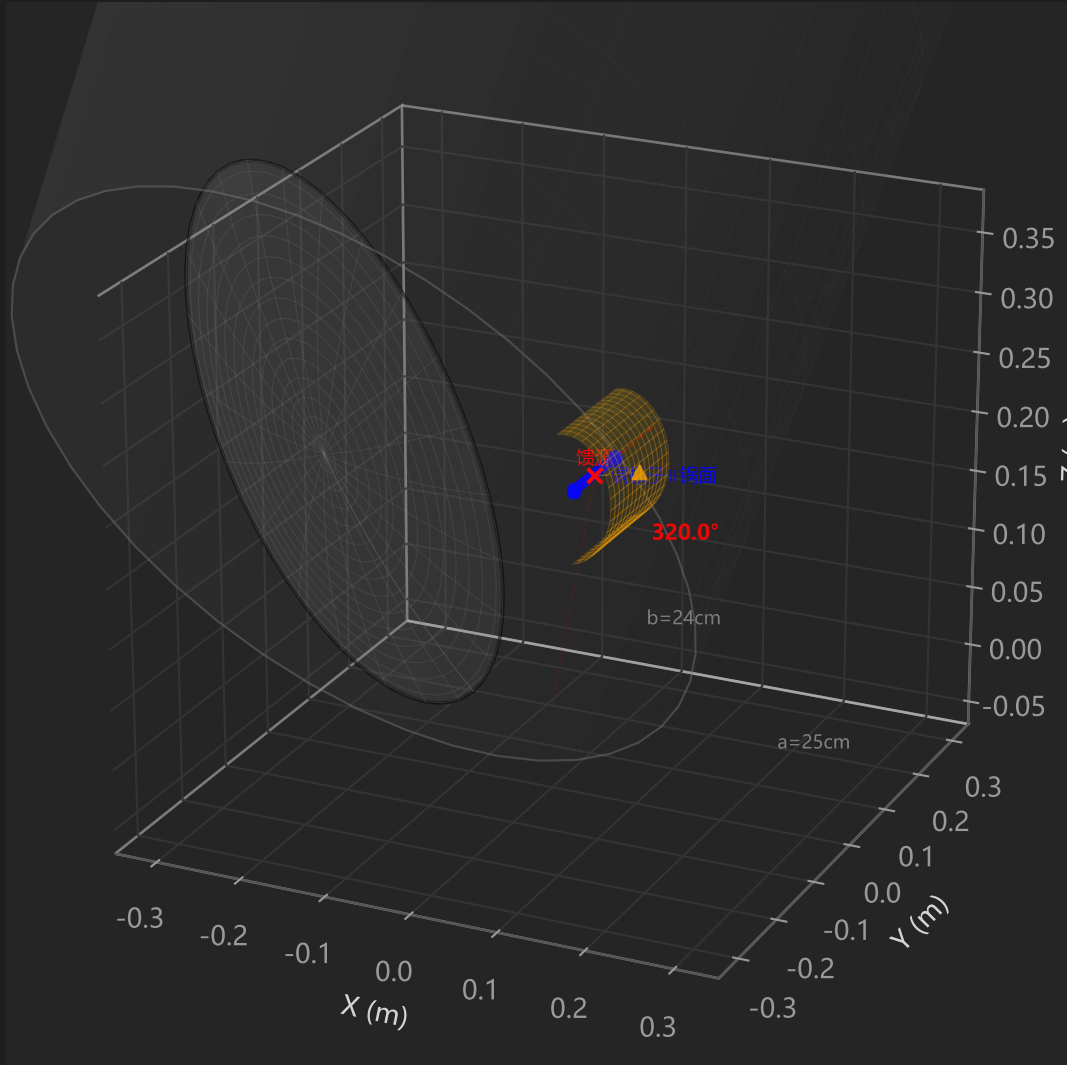
极坐标方向图



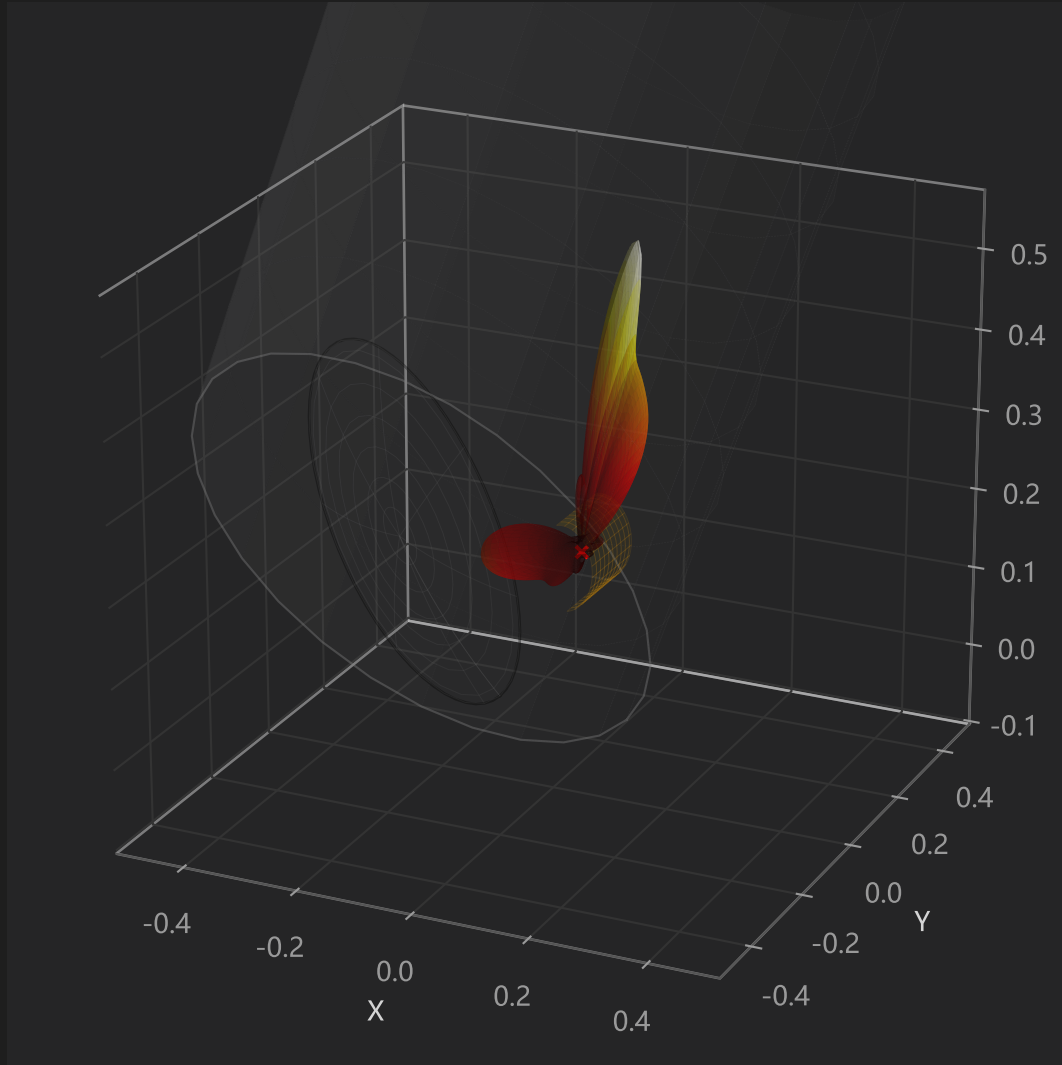
3D 方向图



天线结构



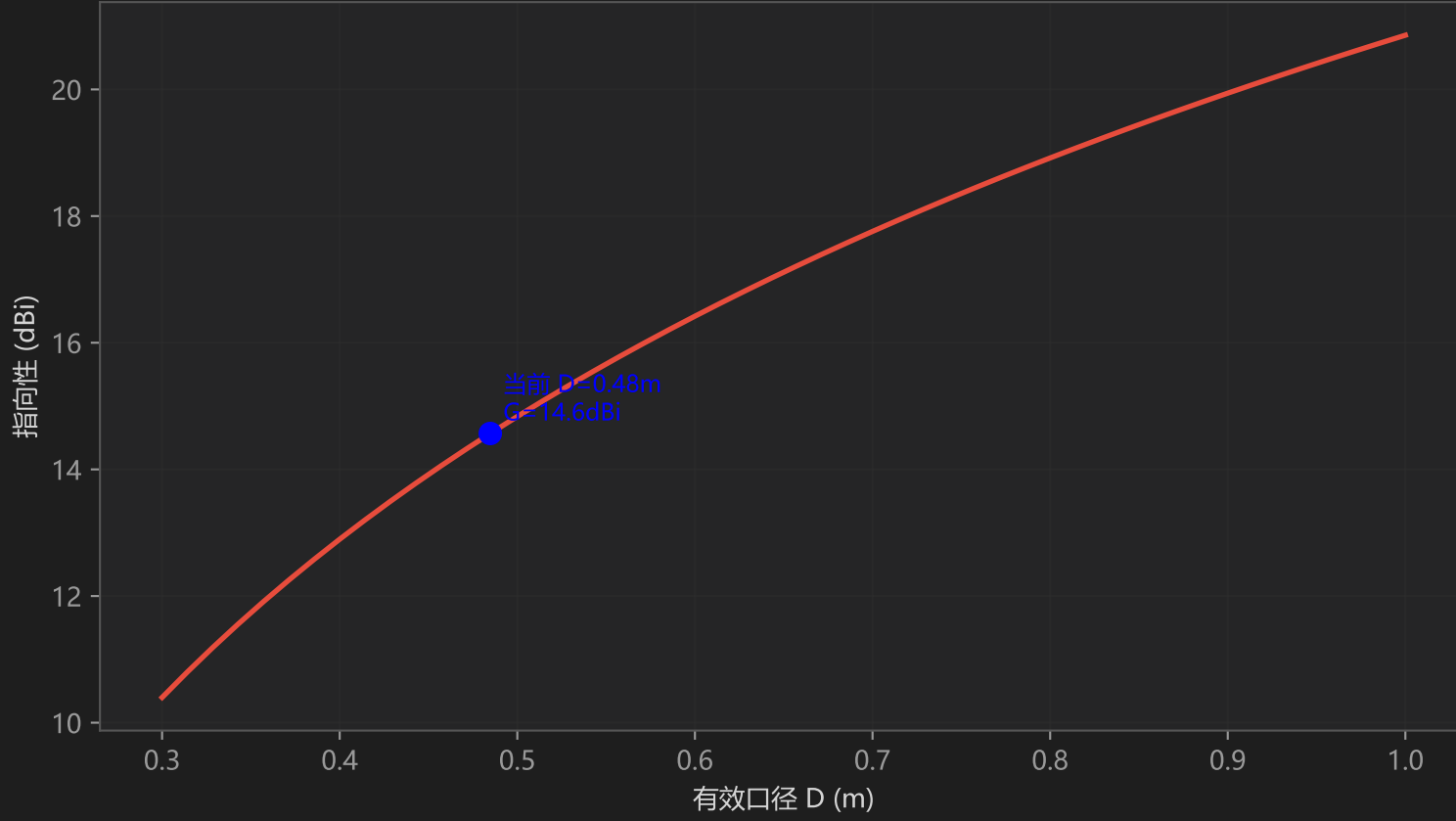
结构 + 方向图叠加



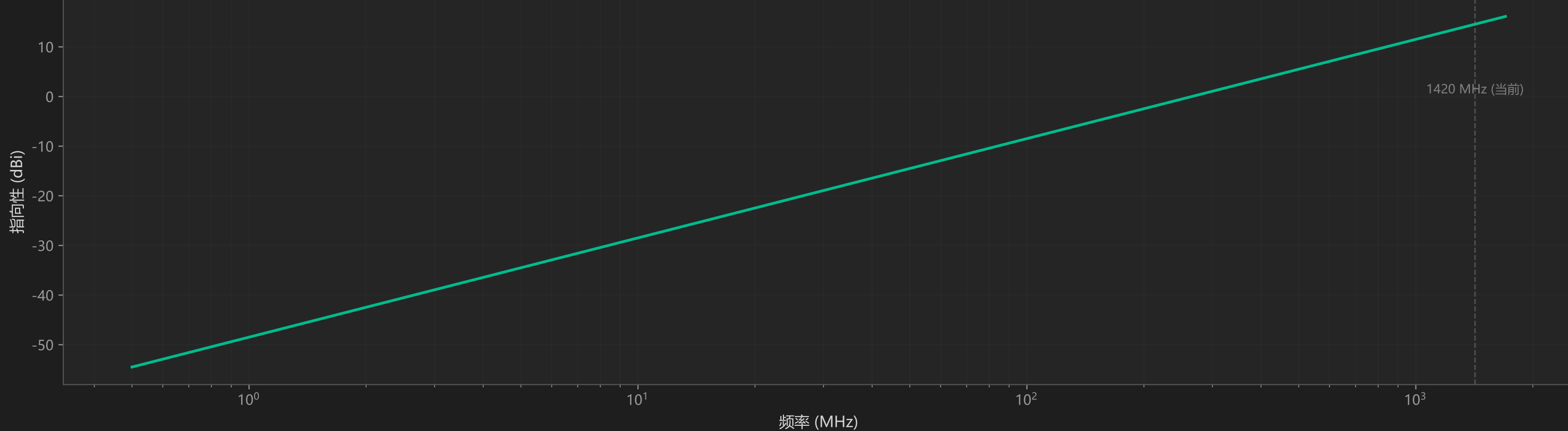
频率: 1420.0 MHz $\lambda=21.1\text{ cm}$
偶极子: 半波长 ($\lambda/2$) $L=105.6\text{ mm}$
锅: $a=25.0\text{cm}$ $b=23.5\text{cm}$ $f=40.0\text{cm}$
偏馈角: 320.0° 反射板距离: 5.3cm
桶: $D=80.0\text{cm}$ $H=100.0\text{cm}$
旋转: $X=0.0^\circ$ $Y=-28.0^\circ$

□ 指向性: 14.57 dBi
□ E面 HPBW: 15.6°
□ H面 HPBW: 29.9°
□ 前后比 F/B: 6.3 dB
□ 有效口径: 0.485m $f/D=0.825$

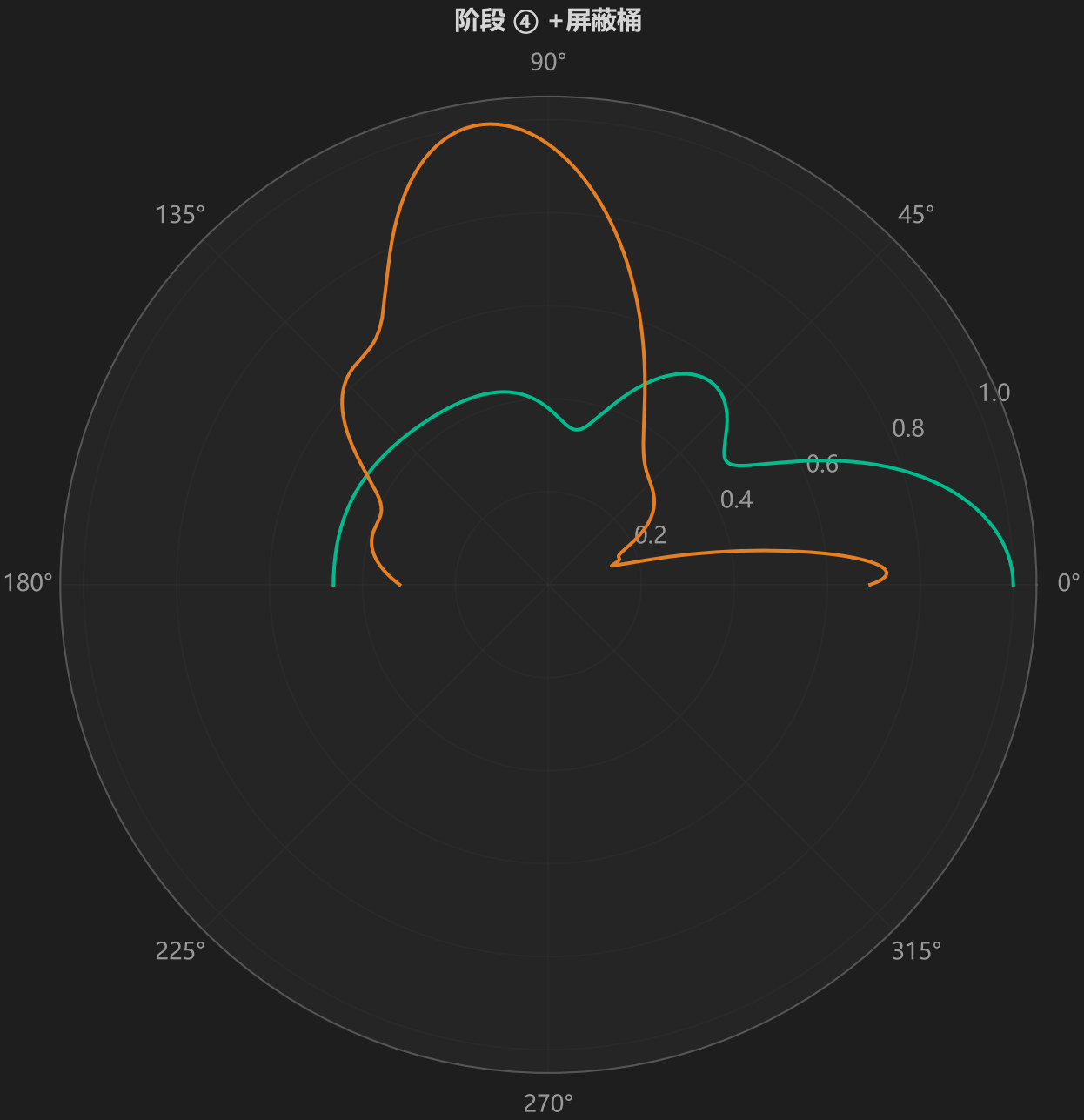
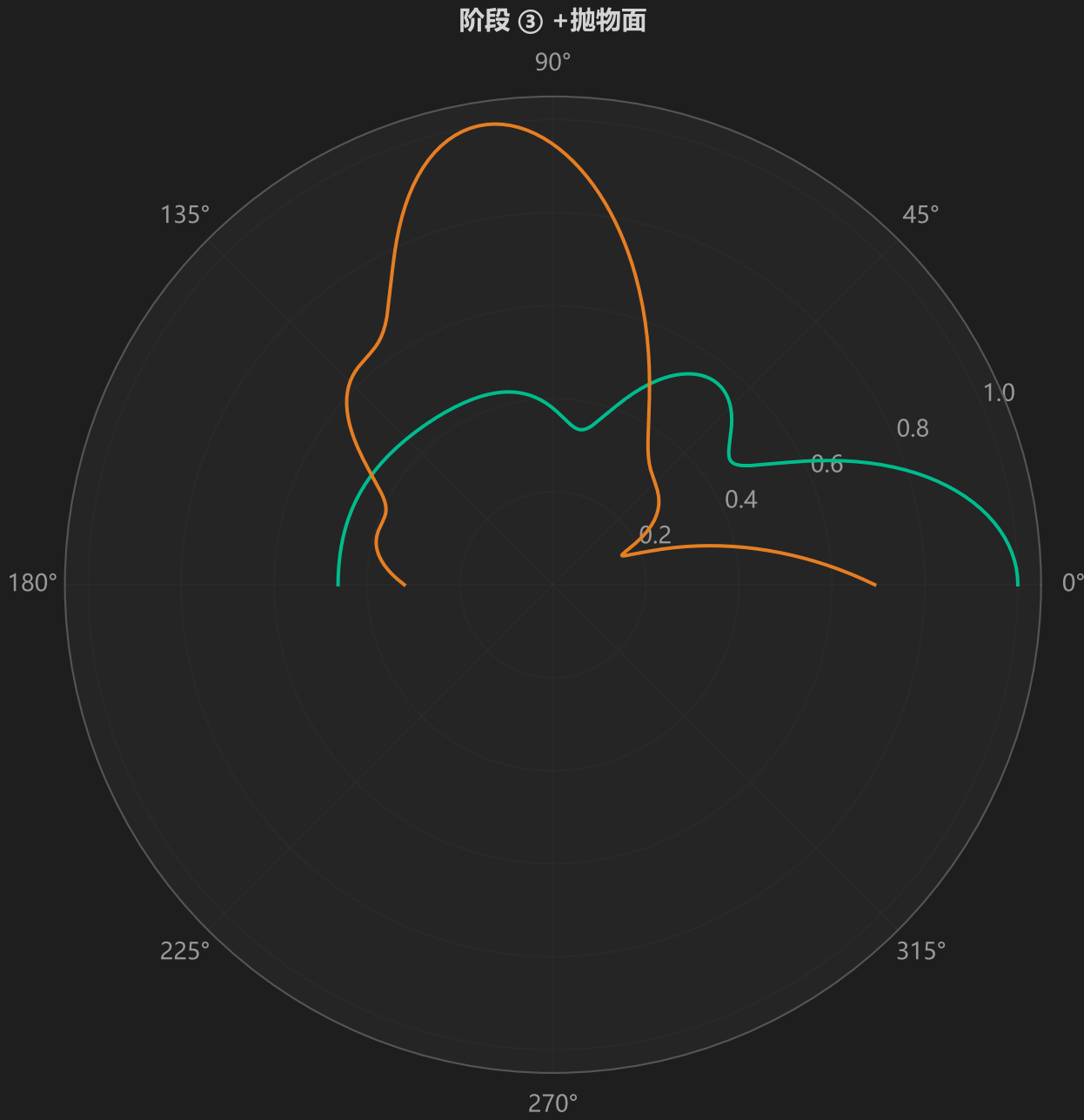
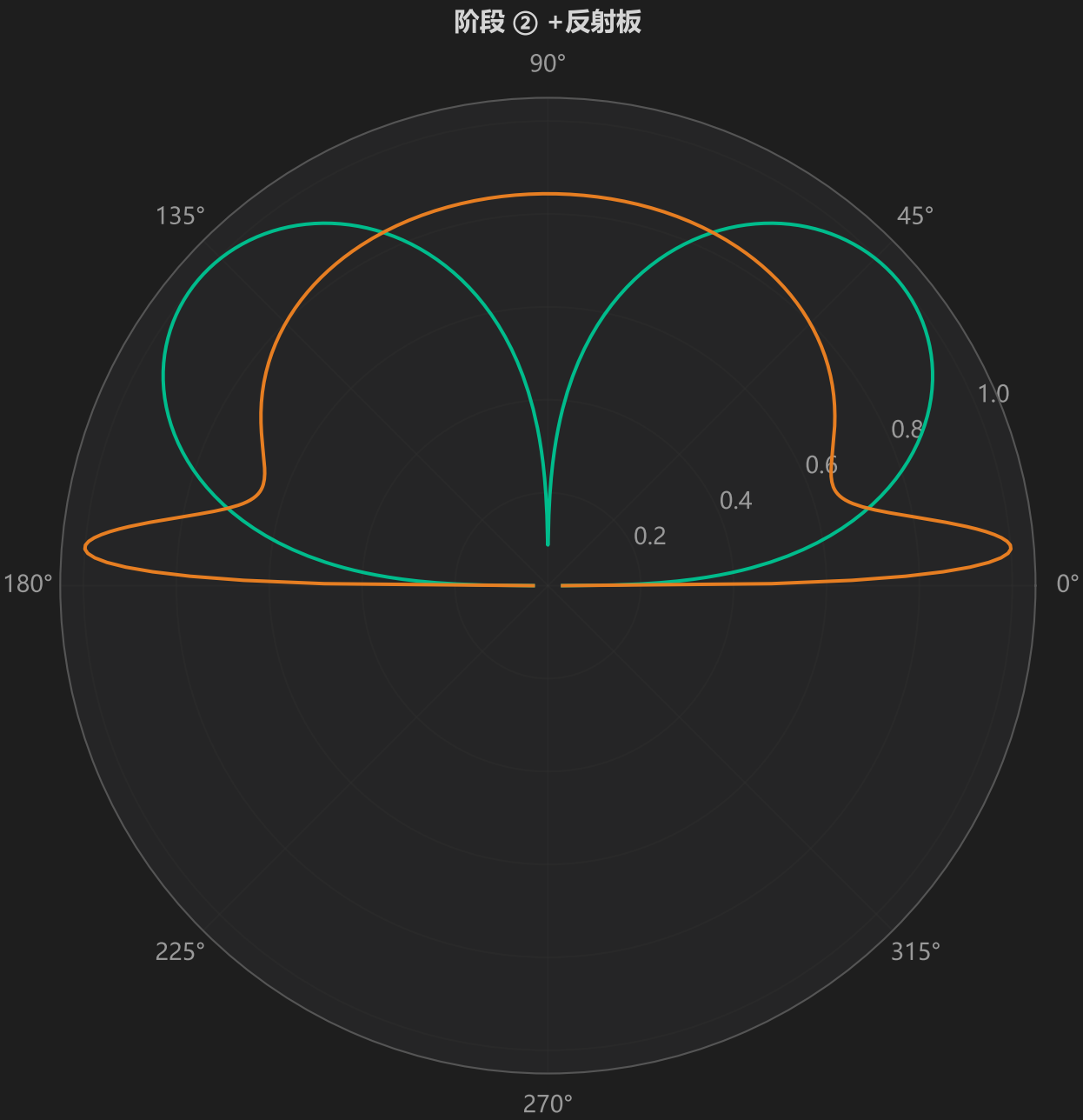
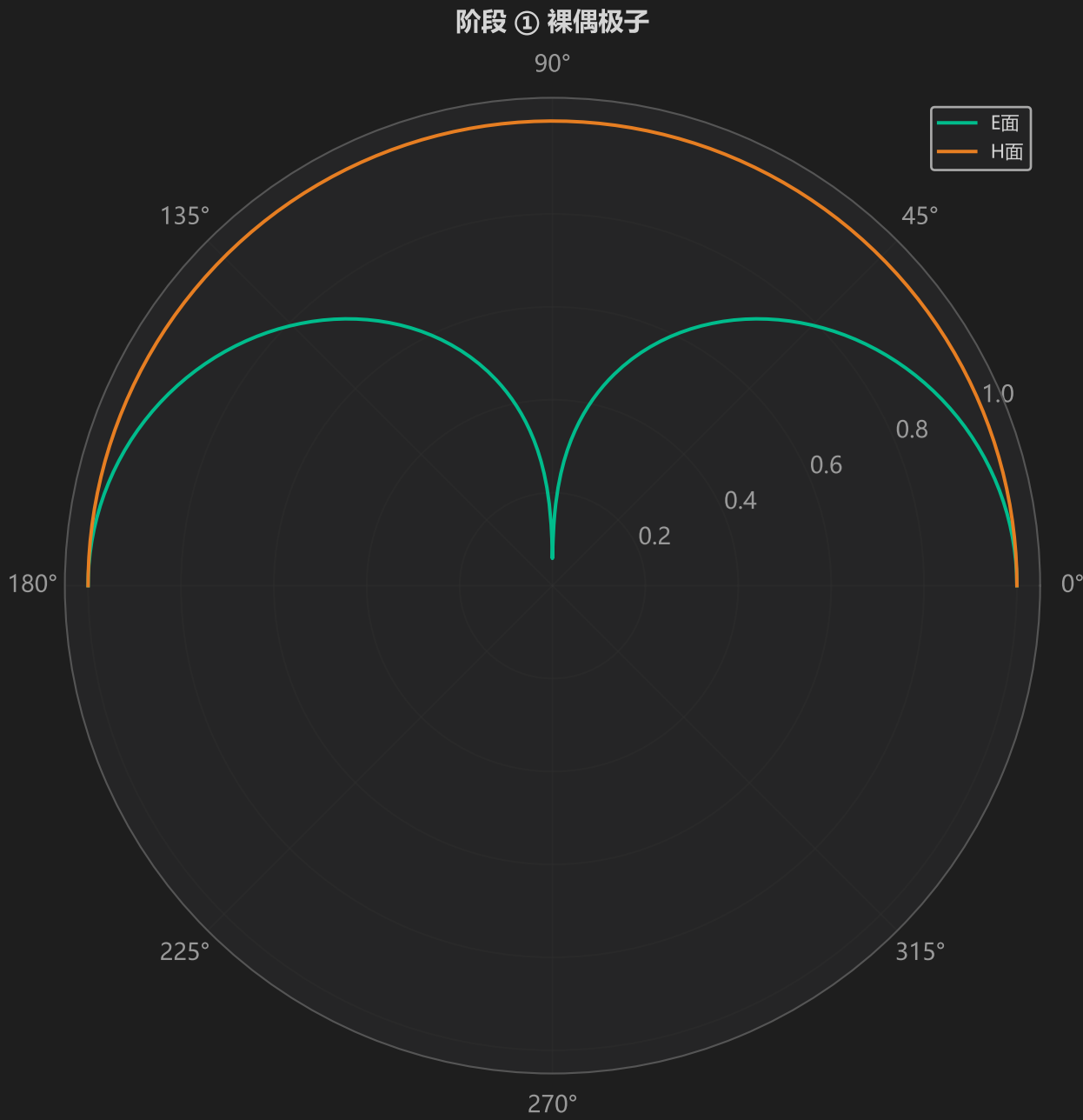
增益 vs 口径 (理论极限)



频率 vs 最大增益 / 指向性 (理论极限)



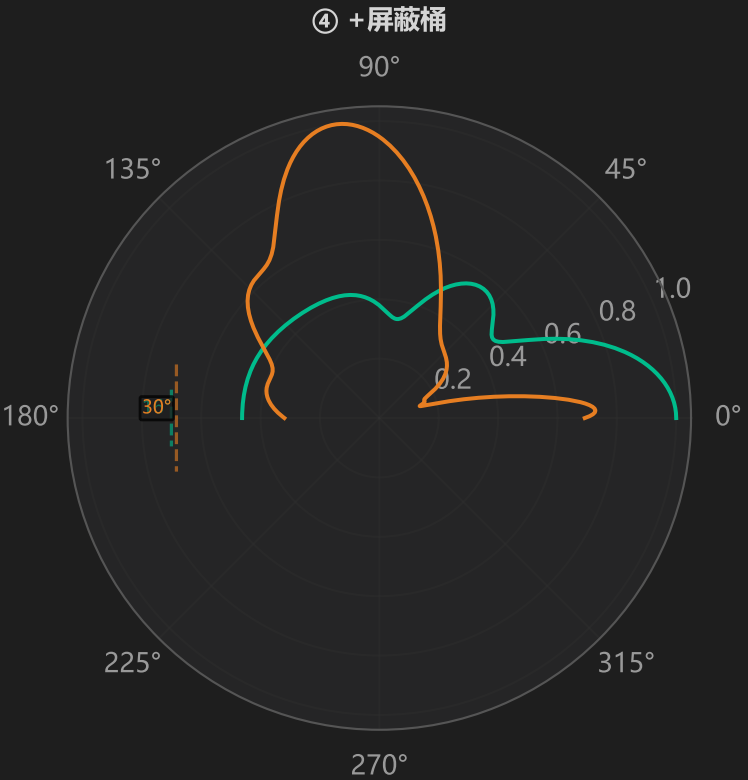
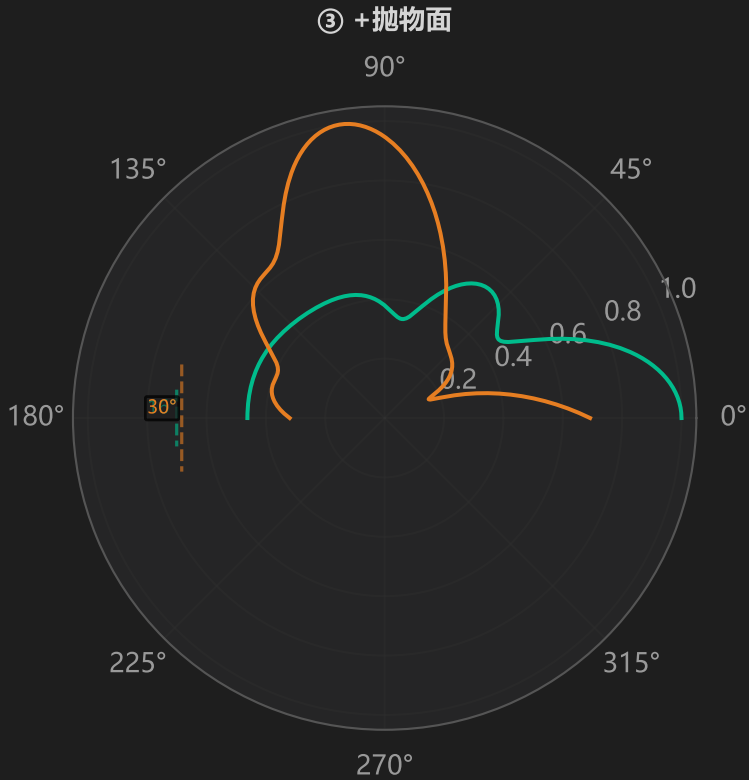
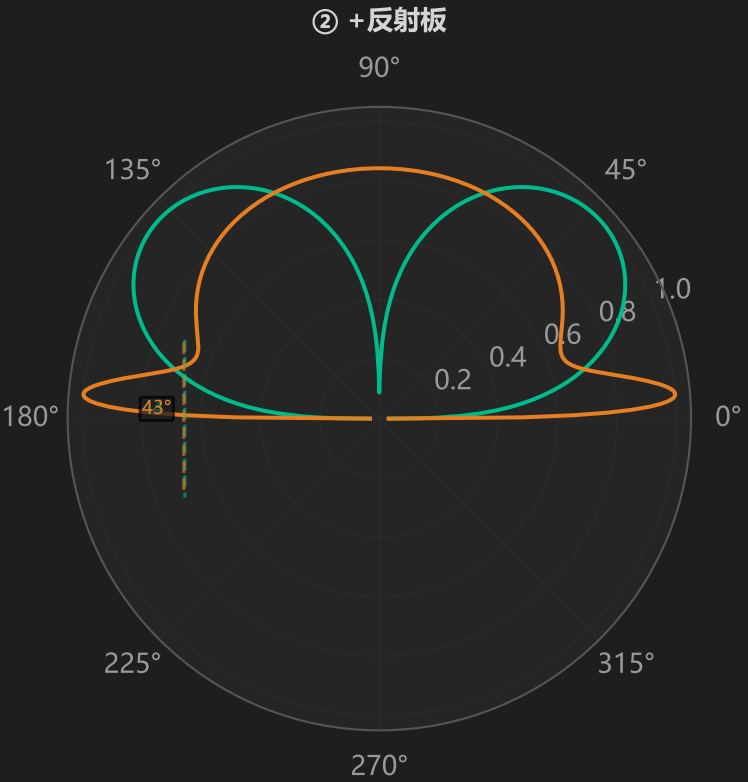
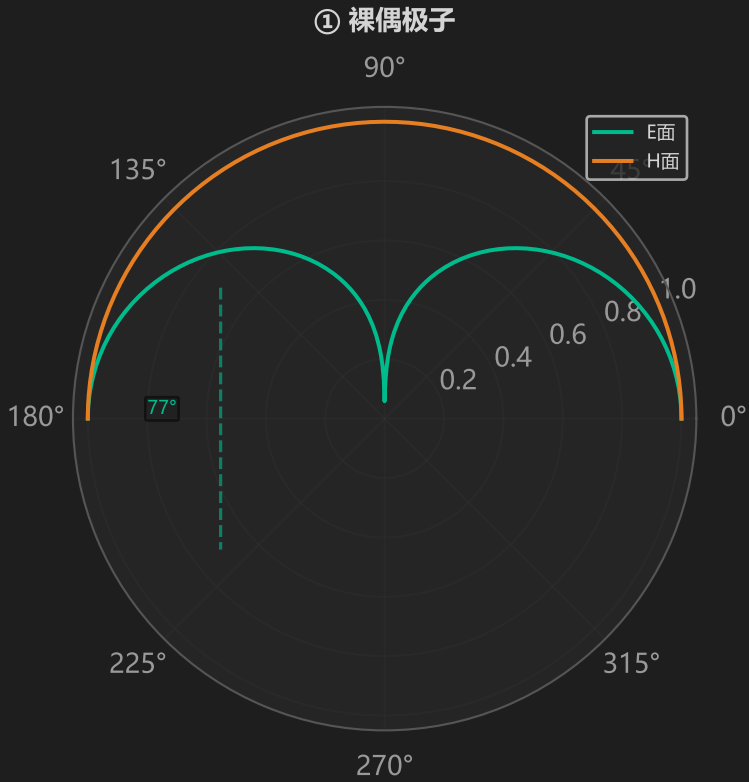
1420.0 MHz 天线仿真 — 渐进结构分析 | 半波长 ($\lambda/2$)



阶段指标演进

阶段	D(dBi)	HPBW_E	HPBW_H	F/B(dB)
① 裸偶极子	1.9	77.2°	180.0°	0.0
② +反射板	3.6	44.1°	42.6°	-17.0
③ +抛物面	4.9	15.5°	29.6°	6.0
④ +屏蔽桶	4.9	15.5°	29.6°	6.0

逐级优化 — 各阶段独立方向图 (无跨级叠加,渲染更轻量)



阶段	D (dBi)	HPBW_E	HPBW_H	F/B (dB)
① 裸偶极子	1.9	77.2°	180.0°	0.0
② +反射板	3.6	44.1°	42.6°	-17.0
③ +抛物面	4.9	15.5°	29.6°	6.0
④ +屏蔽桶	4.9	15.5°	29.6°	6.0